OBI Robotics — Dexterity via Fourier Series

A formal basis for the self-inscribing robot hand

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1. Problem statement

A Fourier-series drawing machine (the epicycle construction: nested circles spinning at angular rates k-omega, tip-to-tail) moves its pen along a parametric curve. The claim to be formalized is threefold:

1. **(Traceability)** Any continuous closed curve in R^3 can be traced to arbitrary precision by a finite chain of rotating elements (epicycles in 2D; elliptical gimbals in 3D).
2. **(Sufficiency)** A 27-DOF humanoid hand carries enough independently controllable cyclic motion to trace its own outline (and hence to print/draw itself) to engineering tolerance.
3. **(Self-inscription)** Self-drawing is a fixed-point property: the manipulator's motion encodes the manipulator's own geometry, gamma* = iota(hand).

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2. Theorem 1 (Traceability of closed curves in R^3)

Statement. Let gamma : [0, 2pi] -> R^3 be continuous with gamma(0) = gamma(2pi). For every eps > 0 there exists N and vectors a_0, a_k, b_k in R^3 (k = 1..N) such that

`gamma_N(t) = a_0 + sum_{k=1..N} [ a_k cos(kt) + b_k sin(kt) ]`
satisfies sup_t ||gamma(t) - gamma_N(t)||_inf < eps.

Proof. Apply the construction coordinatewise and complexify: z_j(t) = gamma_j(t) for the j-th coordinate is a continuous 2pi-periodic function C -> C (identify the codomain axis with the imaginary axis). By the Weierstrass approximation theorem for trigonometric polynomials

- constructively, by **Fejer's theorem**: the Cesaro means of the Fourier partial sums converge uniformly to any continuous periodic function - there is a trigonometric polynomial z_{j,N}(t) = sum_{|k|<=N} c_{j,k} e^{ikt} with sup|z_j - z_{j,N}| < eps. Stacking the three coordinates gives gamma_N; a_k = 2Re(c_k), b_k = -2Im(c_k) termwise. []

Corollary (rate of convergence). If gamma is C^p and its p-th derivative is absolutely continuous (a piecewise-smooth CAD outline satisfies this), then |c_k| = O(|k|^-(p+1)); for the smoothed outline measured in sec.7 the observed decay is consistent with the corner-limited (p = 1) regime: sup-error falls as ~ C/N^2 asymptotically, and in practice exponentially until the corner floor is reached.

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3. Geometric reading: circles become gimballed ellipses in 3D

Lemma (ellipse). For alpha, beta in R^3 not both zero, the image of E(t) = alpha cos t + beta sin t is an ellipse in the plane span{alpha, beta}, with semi-axes (|alpha|+|beta|)/2 and ||alpha|-|beta||/2 (degenerate cases: a line segment when alpha || beta, a point when alpha = beta = 0). It lies on the sphere of radius (|alpha|+|beta|)/2 about the attachment point.

Proof. Write E(t) = Re[(alpha - i beta)e^{it}]; in the orthonormal basis of the plane the map is a diagonal linear image of the unit circle. []

So the 3D successor of the 2D circle chain - the "overlapping spheres" of the design notes - is a **tip-to-tail chain of elliptical orbits**, each riding on a sphere nested inside its parent's sphere. Each harmonic k contributes:

- **2 orientation DOF** - the plane normal n = (alpha x beta)/|alpha x beta|: *roll* and *pitch* (right-hand rule fixes the sign convention; reversing k interchanges clockwise / counter-clockwise spin);
- **1 spin DOF** - rotation about n at rate k-omega: *yaw*.

This is exactly the roll-pitch-yaw decomposition sketched in the notes, with the right-hand rule assigning the sense of orbit per harmonic. The chain is continuous by construction (tip-to-tail summation), so the pen never lifts - tracing, not stamping.

Kinematic mapping.

Actuation scheme	DOF per harmonic	27 DOF buys	Tracing error (measured, 3D)
Gimbal-actuated (roll+pitch+yaw all actuated)	3 (+3 base)	N = 8 harmonics	3.87 units = 2.15% of hand length
Spin-actuated (planes fixed at calibration)	1 (+1 base)	N = 26 harmonics	0.39 units = 0.22% of hand length
Tolerance +-2.0 units (1.1% of L)	-	gimbal: N >= 14	met
Tolerance +-1.0 unit (0.56% of L)	-	gimbal: N >= 19	met

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4. The 27-DOF accounting

The standard humanoid-hand count (matches the 27 assumed in the notes):

Segment	DOF	Detail
Four fingers	4 x 4 = 16	MCP: 2 flexion + 1 abduction; PIP: 1; DIP: 1
Thumb	5	CMC: 2 (saddle), MCP: 1, IP: 2 (twist + flexion)
Wrist	2	flexion/extension, radial/ulnar deviation
Forearm	2	pronation/supination, axial translation lock-out
Palm arch	2	transverse + longitudinal metacarpal curvature
Total	**27**	

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5. Dexterity: a composite, measurable index

Dexterity is decomposed into four measurable factors, normalized to [0,1] and combined geometrically (a chain is as dextrous as its weakest link):

1. **Kinematic bandwidth** D1 = min(1, N_available / N*(eps)) - harmonics realizable vs. harmonics demanded by the task at tolerance eps (this document's central quantity).
2. **Manipulability** (Yoshikawa): w(q) = sqrt(det J(q)J(q)^T) for task Jacobian J; D2 = w/w_max.
3. **Isotropy**: D3 = sigma_min(J)/sigma_max(J) - directional uniformity of the velocity ellipsoid.
4. **Workspace coverage**: D4 = Vol(W intersect Gamma)/Vol(Gamma) - fraction of the task geometry reachable.

Composite: **D = (D1 D2 D3 D4)^{1/4}**. A hand that cannot reach its own outline has D4 = 0 and therefore D = 0 regardless of joint count - reach precedes dexterity.

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6. The self-inscription fixed point

Definition. A manipulator M with workspace W(M) and hull boundary Sigma(M) is *self-inscribing* iff Sigma(M) is contained in W(M).

Theorem 2 (self-printing). If M is self-inscribing and Sigma(M) is a piecewise-C^1 closed curve (or a piecewise-smooth closed surface, reduced to its boundary curves), then M can trace its own outline: by Theorem 1 the boundary curve admits a finite Fourier-epicycle parameterization realizable within the kinematic bandwidth budget of sec.5.

Proof. Sigma(M) piecewise-C^1 implies rectifiable and closed, so gamma(t) is continuous and piecewise smooth. Theorem 1 applies with Fourier coefficients computable by direct measurement of Sigma(M) - the hand is both the measuring instrument and the measured object. []

Fixed-point reading. Let iota denote the geometric-embedding map (hand -> its outline) and tau the trajectory-encoding map (outline -> joint trajectories). Self-inscription is the commuting condition tau(iota(M)) = program(M): *the program that draws the hand is derived from the hand itself.* Because the workspace of a larger hand covers the outline of a smaller one, one master hand can print arbitrary copies at reduced scale - a manufacturing cascade seeded by a single self-inscribing instance.

Honest caveat. A real finger does not spin like an epicycle; the Fourier construction is a *trajectory encoding*, and joint-space realization proceeds through the Jacobian: q-dot = J^+(q) gamma-dot_N(t). The epicycle chain is the canonical flat output; the hand's 27 DOF are the actuation budget that must carry it. D1 is precisely the measure of whether the budget suffices - and per the measurements in sec.7, it does.

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7. Measured results (this build)

Hand outline: parametric silhouette, L = 179.9 units (wrist to middle fingertip), 512-point periodic resampling, periodic Gaussian smoothing sigma = 8 (the physical outline is smooth; rasterization is not). Fourier coefficients by direct FFT; error = sup-norm tracing error.

N harmonics	2D error	3D error	% of L (3D)
1	86.37	81.96	45.56%
2	71.39	70.37	39.11%
4	40.27	40.24	22.37%
6	15.79	15.67	8.71%
8	4.60	3.87	2.15%
12	3.37	3.20	1.78%
16	1.40	1.34	0.74%
20	0.59	0.55	0.31%
24	0.37	0.36	0.20%
26	0.40	0.39	0.22%
32	0.04	0.03	0.02%
48	0.00	0.00	0.00%
64	0.00	0.00	0.00%

- **Tolerance +-2.0 units (1.1% of L):** N* = 14 harmonics (2D and 3D agree).
- **Tolerance +-1.0 unit (0.56% of L):** N* = 19 harmonics.
- **27 DOF, gimbal scheme (N = 8):** 3.87 units - *the hand traces its own outline to 2.15% of its length using under a third of its kinematic budget.*

- **27 DOF, spin scheme (N = 26):** 0.39 units - sub-millimeter scale at hand scale.

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8. Conclusion

The intuition in the design notes is correct and now has a proof skeleton: the epicycle construction is a universal curve-tracing mechanism in R^3 (Theorem 1), its 3D form is a roll-pitch-yaw gimbal chain of elliptical orbits riding nested spheres (Lemma, sec.3), a 27-DOF hand's kinematic budget is *sufficient* to trace its own outline to engineering tolerance (measured, sec.7), and self-printing is the fixed point tau o iota = id (Theorem 2). Dexterity is therefore not a vague notion but a bandwidth number: **the number of independent cyclic motions a manipulator can command, measured against the harmonic content of the shape it must draw.**